

01_Data_Structure_SegTree线段树板子

[P3374 【模板】树状数组 1 - 洛谷](#)

```
#include<bits/stdc++.h>
#define lp (p << 1)
#define rp (p << 1 | 1) //?等效于 #define rp (p << 1 + 1)
using namespace std;
using ll = long long;
constexpr int inf = 0x3f3f3f3f;

struct SegTree{ //?
    int n;
    vector<ll> a;
    vector<ll> sum;

    //构造, 初始化函数
    void init(int n_){
        n = n_;
        sum = vector<ll>((n+1) << 2);
    }

    void init(int n_, vector<ll>& a_){
        n = n_;
        a = a_;
        sum = vector<ll>((n+1) << 2);
        build(1, 1, n);
    }

    void pull(int p){
        sum[p] = sum[lp] + sum[rp];
    }

    void build(int p, int l, int r){
        if(l == r){
            sum[p] = a[l];
            return;
        }
        int mid = (l+r) >> 1;

        build(lp, l, mid);
        build(rp, mid+1, r);
        pull(p);
    }

    //单点修改
    void update(int p, int l, int r, int x, ll v){
        if(l == r){
            sum[p] += v;
            return;
        }
    }
```

```

        int mid = (l+r) >> 1;
        if(x <= mid) update(lp, l, mid, x, v);
        else update(rp, mid+1, r, x, v);
        pull(p);
    }

    //区间查询
    ll query(int p, int l, int r, int x, int y){
        if(x <= l && r <= y) return sum[p];

        ll ans = 0;
        int mid = (l+r) >> 1;

        if(x <= mid) ans += query(lp, l, mid, x, y);
        if(y >= mid+1) ans += query(rp, mid+1, r, x, y);

        return ans;
    }

    //相当于对外接口，调用更简便
    void update(int x, ll v){ update(1, 0, n, x, v);} //单点修改
    ll query(int L, int R){ return query(1, 0, n, L, R);} //区间查询
};

void work(){
    int n,q;
    cin >> n >> q;
    vector<ll> a(n+1);
    for(int i=1; i<=n; i++){
        cin >> a[i];
    }
    SegTree sgt;
    sgt.init(n,a);

    vector<int> ans;
    for(int i=1; i<=q; i++){
        int op;
        cin >> op;
        if(op == 1){
            int x, k;
            cin >> x >> k;
            sgt.update(1, 1, n, x, k);
        }
        else if(op == 2){
            int x, y;
            cin >> x >> y;
            ll anss = sgt.query(1, 1, n, x, y);
            cout << anss << '\n';
        }
    }
}

signed main(){

```

```
ios::sync_with_stdio(0);
cin.tie(0);

int t=1;
// cin >> t;
while(t-->0) work();
}

/*
5 5
1 5 4 2 3
1 1 3
2 2 5
1 3 -1
1 4 2
2 1 4

14
16
*/
```